# EUGENE LEE

PhD Candidate in Econometrics | Korea University | eugenelee1008@gmail.com

## RESEARCH INTERESTS

Natural Language Processing (NLP) · Treatment Effects · Machine Learning · Applied Econometrics · Environmental Economics

## EDUCATION

**Ph.D. in Econometrics** | Korea University | Expected 2026

**M.A. in Economics** | Korea University | 2020

**Bachelor of Commerce** | Australian National University | 2017

## PEER-REVIEWED JOURNAL PUBLICATIONS

**Lee, Eugene**, Myoung-jae Lee, and Goeun Lee (2026). "Extending policies to control groups or past periods in nonlinear difference in differences." *Applied Economics*, 58(45), 9949–9962.

DOI: https://doi.org/10.1080/00036846.2025.2550668

Myung, Rangmi*, **Eugene Lee***, Jinyeong Kim, Jieun Kim, and Hyunjoo Pai (2025). "Attributable costs of Clostridioides difficile infections in Korea." *Journal of Korean Medical Science*, 40(4), e22.

* Equal first authors. DOI: https://doi.org/10.3346/jkms.2025.40.e22

## DISSERTATION RESEARCH — WORKS IN PROGRESS

**NLP-Based Analysis of Public Attitudes Toward Climate Change in Korea** *(Chapter 1; working title)*

- Analyzes approximately 50,000 Korean YouTube comments using LDA topic modeling and a fine-tuned KoELECTRA sentiment classifier, with manually labeled data and model-validation procedures.

**Evaluating California's Cap-and-Trade Program Using Facility-Level Emissions** *(Chapter 2; working title)*

- Empirical analysis of greenhouse-gas and co-pollutant emissions, with attention to regulatory coverage, facility transitions, and causal identification.

**Difference-in-Differences with Changing Treatment Qualification** *(Chapter 3; working title)*

- Develops stayer-based DiD and conditional response function approaches for time-varying treatment qualification; evaluates estimators through Monte Carlo simulations and a California application.

## RESEARCH EXPERIENCE

**Korea Development Institute (KDI)** | Researcher | 2021–2023

- Supported policy research on carbon taxation, emissions trading, low-carbon transition, and renewable energy markets, including data collection and organization for KDI reports.

## SELECTED KDI POLICY REPORTS — RESEARCH ASSISTANCE

Yoon, Yeochang (2023). *Renewable Energy Policy in Korea* [재생에너지 구매제도 및 지원정책 개선 과제]. KDI Policy Study Series No. 2023-04. Research assistance acknowledged.

Yang, Yong Hyeon (ed.) (2023). *Green Transition Strategy and Implementation* [저탄소경제 전환 전략과 정책과제]. KDI Research Monograph No. 2023-02. Credited research support.

Yoon, Yeochang (2021). *Carbon Taxation Policy in Korea* [탄소세 도입 방안에 대한 연구]. KDI Policy Study Series No. 2021-08. Research assistance acknowledged.

## TECHNICAL SKILLS

**Programming / analysis:** R, Python; econometric modeling, statistical simulation, NLP, topic modeling, transformer-based text classification.